PYTHON / QUANTITATIVE ENERGY ANALYTICS

Power Portfolio Analytics

Positions, wholesale cash flows and scenario comparison

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Independent analytical software project

Illustrative portfolio | One ordinary 24-hour day | 48 half-hour periods

One portfolio, evaluated period by period

Generation and customer consumption determine the pre-trade exposure.

ELECTRICITY GENERATION

Wind + solar

Total energy in each half-hour, MWh

−

SUPPLY OBLIGATION

Customers

Consumption in the same period, MWh

=

SIGNED POSITION

Long / short

Before wholesale purchases or sales

Shortage → purchase | Surplus → sale

Period cash flow = sale quantity × price − purchase quantity × price

One supplied price per period. Negative prices are valid. No storage or inter-period energy transfer.

One calculation engine, two analytical interfaces

File-based inputs and modular responsibilities keep the comparison consistent.

READ

CSV inputs
Date + 48 records

→

ORGANISE

PortfolioDay
Period checks

→

CALCULATE

Positions and trades
Daily aggregation

→

REPORT

Console + CSV
Four PNG charts

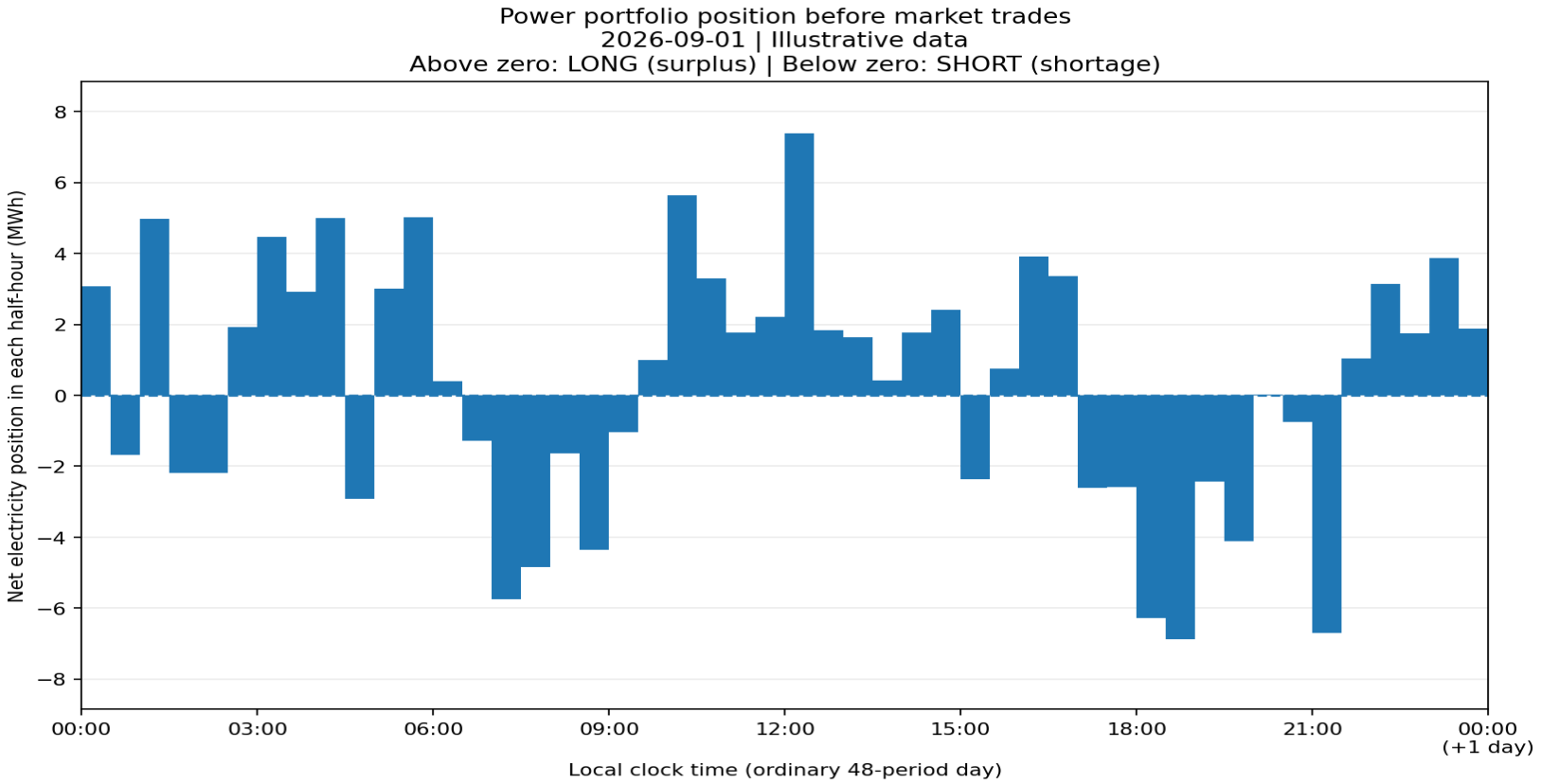
SCENARIO COMPARISON

Baseline and alternative CSVs → same engine → metric-by-metric differences

Difference = scenario − baseline. Same delivery date and complete 48-period coverage required.

Timing matters, even with a daily energy surplus

Synthetic inputs for 1 September 2026. Bars show positions before market trades.



PERIOD COUNTS

29 / 19

Long / short; 0 balanced periods

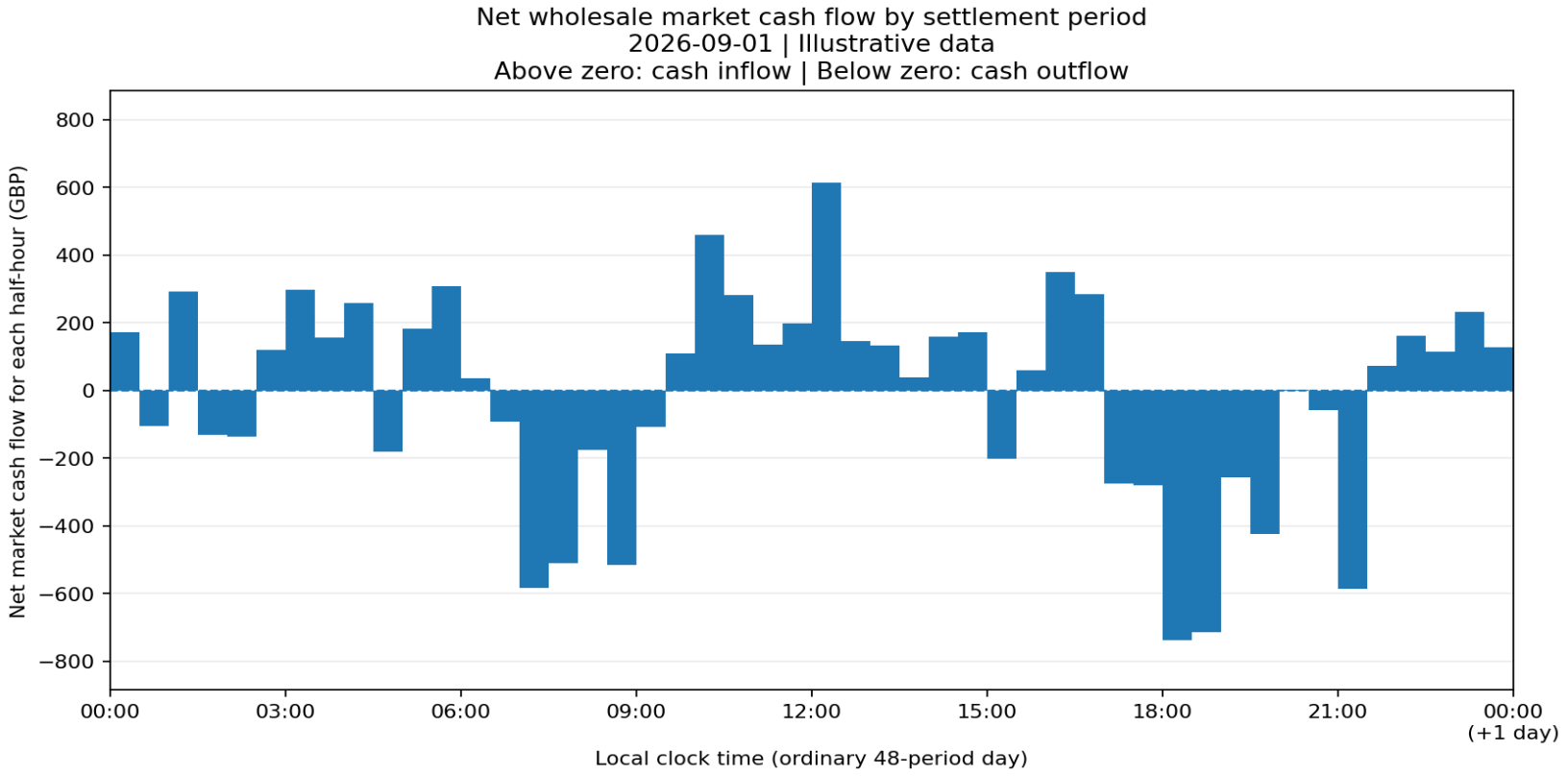
Largest surplus

+7.38 MWh

Period 25, 12:00–12:30

More MWh sold does not guarantee a cash inflow

Both the traded quantity and the applicable price determine the monetary result.



PURCHASES

62.67 MWh

£97.10/MWh volume-weighted

SALES

79.81 MWh

£70.77/MWh volume-weighted

NET MARKET CASH FLOW

−£437.28

A controlled input change gives an explainable result

Increase period 1 consumption by 1 MWh. Keep all other inputs unchanged.

Daily metric	Baseline	Scenario	Difference
Consumption (MWh)	554.67	555.67	**+1.00**
Purchases (MWh)	62.67	62.67	**0.00**
Sales (MWh)	79.81	78.81	**−1.00**
Market cash flow (GBP)	−437.28	−492.78	**−55.50**

Period 1 remains long: 1 MWh less sold × £55.50/MWh = £55.50 lower cash flow.

Select files from the command line

The same calculations support an individual report and a two-case comparison.

SINGLE CASE

```
python main.py --input data/example.csv --output-dir outputs/run_a
```

TWO CASES

```
python compare_scenarios.py --baseline data/base.csv
  --scenario data/alternative.csv --output outputs/comparison.csv
```

Illustrative command paths. Exact reference-file commands and environment setup are in README.md.

CSV inputs remain unchanged. Repeated runs replace selected outputs, so use distinct destinations.

Checks support, but do not replace, interpretation

The reference snapshot contains 30 automated tests. Coverage is targeted, not exhaustive.

FINANCIAL EXPECTATIONS	Fixed cases establish quantities and cash-flow signs independently of the model.
INTERFACE AND FILES	Checks cover argument forwarding, period coverage, exports and input-file protection.
SCENARIO CONSISTENCY	Identical inputs, reversed comparisons and controlled changes have known outcomes.

Modular Python • pytest • Ruff development checks • shared agent instructions • Git version history

A defined analytical model, not a trading platform

The limitations are part of the result and should remain visible with the code.

IMPLEMENTED

One wind farm and one solar PV farm
Customer supply obligation
48-period position and cash-flow analysis
CSV inputs and numerical / visual outputs
Controlled two-scenario comparison

NOT REPRESENTED

Live execution or imbalance settlement
Forecasting, storage or optimisation
PPAs, forwards or hedge positions
Retail revenue and full operating costs
Clock-change days or network constraints

Explore the code, reference inputs, charts and comparison results:

github.com/spgian/power_portfolio